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Assessment of the October 2024 Cut-Off Low Event Floods Impact in Valencia (Spain) with Satellite and Geospatial Data

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Abstract

The October 2024 cut-off low event triggered one of the most catastrophic floods recorded in the Valencia Metropolitan Area, exposing significant vulnerabilities in urban planning, infrastructure resilience, and emergency preparedness. This study presents a novel comprehensive assessment of the event, using a multi-sensor satellite approach combined with socio-economic and infrastructure data at the metropolitan scale. It provides a comprehensive spatial assessment of the flood's impacts by integrating of radar Sentinel-1 and optical Sentinel-2 and Landsat 8 imagery with datasets including population density, land use, and critical infrastructure layers. Approximately 199 km^2 were inundated, directly affecting over 90,000 residents and compromising vital infrastructure such as hospitals, schools, transportation corridors, and agricultural lands. Results highlight the exposure of peri-urban zones and agricultural areas, reflecting the socio-economic risks associated with the rapid urban expansion into flood-prone plains. The applied methodology demonstrates the essential role of multi-sensor remote sensing in accurately delineating flood extents and assessing socio-economic impacts. This approach constitutes a transferable framework for enhancing disaster risk management strategies in other Mediterranean urban regions. As extreme hydrometeorological events become more frequent under changing climatic conditions, the findings underscore the urgent need for integrating remote sensing technologies, early warning systems, and nature-based solutions into regional governance to strengthen resilience, reduce vulnerabilities, and mitigate future flood risks.

Keywords: cut-off low; DANA; Valencia; flood mapping; Sentinel-1; Sentinel-2; Landsat 8; Geographic Information System; Critical Infrastructures; exposure assessment; natural hazards

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Academic Editor: Marios Anagnostou

Received: 29 March 2025
Revised: 23 May 2025
Accepted: 20 June 2025
Published: 22 June 2025

Citation: Castro-Melgar, I.; Falaras, T.; Basiou, E.; Parcharidis, I. Assessment of the October 2024 Cut-Off Low Event Floods Impact in Valencia (Spain) with Satellite and Geospatial Data. *Remote Sens.* **2025**, *17*, 2145. https://doi.org/10.3390/rs17132145

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1. Introduction

Floods are among the most devastating natural disasters, impacting millions of people worldwide and causing significant economic losses. In Europe, floods have been exacerbated by climate change and increasing urbanization, with Mediterranean coastal regions being particularly vulnerable due to their complex orographic and climatic conditions [1,2]. The frequency and intensity of extreme precipitation events have increased in recent decades, contributing to more severe flooding episodes, particularly in densely populated urban areas [3]. The metropolitan area of Valencia exemplifies these challenges,

as it has experienced recurrent flood events, including the severe flooding caused by the cut-off low event in October 2024 [4].

Cut-off lows are upper-level atmospheric disturbances that often produce intense precipitation and widespread flooding in the western Mediterranean region, especially in autumn. These systems have been responsible for some of the most catastrophic floods in eastern Spain, including those affecting the Valencia region [5]. Previous studies have identified a strong correlation between precipitation induced by cut-off low events and severe flood events in the Iberian Peninsula, with future climate projections suggesting an increase in their intensity due to rising sea surface temperatures and atmospheric instability [2,6].

Llasat et al. (2007) reported that cut-off low systems were linked to 7 out of the 22 most severe flood events in Mediterranean Spain during the period from 1950 to 2000 [7]. Another study indicates that the occurrence of cut-off lows has increased annually on a global scale between 1960 and 2017, with a similar upward trend observed in the European region [8]. Although numerous studies have examined the relationship between floods and urbanization [9–11], there remains a lack of scientific literature that comprehensively analyzes the interplay between population density, critical infrastructure vulnerability, and flood extent, particularly in regions with high recurrence of flood events such as the Valencia Metropolitan Area. Previous efforts in flood risk assessment in the Valencia region have primarily focused on specific sub-regions or particular catchments. Camarasa-Belmonte & Soriano-García [12] developed a hydrogeomorphological methodology to map flood hazards in the Barranco del Carraixet and Rambla de Poyo floodplains, emphasizing the relationship between land use transformations and increasing flood risk in peri-urban environments. Additionally, Eguibar et al. [13] assessed flood hazards in the flat coastal area of Oliva, located south of Valencia, highlighting the combined impact of geomorphological conditions and urbanization on recurrent flood events. However, despite these valuable contributions, there remains a research gap in analyzing the combined effects of population density, critical infrastructure vulnerability, and flood extent at the metropolitan scale.

In recent years, other extreme weather events such, as Mediterranean tropical-like cyclones, commonly known as *Medicanes*, and intense storm systems have also contributed to severe flooding in the broader Mediterranean region. For instance, the *Medicane* "Ianos" in 2020 and Storm Daniel in 2023, both impacting Greece, led to devastating floods, predominately in the Thessaly region. These events resulted in substantial damage to housing, infrastructure, and agriculture, alongside injuries and loss of life. Such cases illustrate the increasing trend in high-impact hydrometeorological phenomena affecting southern European countries and highlight the shared vulnerability of Mediterranean coastal regions to extreme precipitation events [14,15].

Remote sensing has become an essential tool for monitoring and assessing various natural hazards, providing crucial insights into disaster risk management. The ability to acquire frequent, large-scale, and high-resolution data enables scientists and policy-makers to evaluate hazard-prone areas and improve early warning systems. It has been widely applied in the assessment of ground deformation due to seismic activity and volcanic processes, as well as in monitoring wildfires and land subsidence [16–18]. In the case of flood events, remote sensing plays a fundamental role in delineating flood extent, assessing affected infrastructure, and estimating the population exposed to inundation hazards [19,20]. Satellite-based flood mapping provides critical data, especially in data-scarce regions, where traditional ground-based measurements are often insufficient or unavailable [21]. Multispectral and Synthetic Aperture Radar (SAR) imagery allow for flood detection even under cloudy conditions, enhancing real-time response capabilities and supporting insurance mechanisms through flood index models [19]. Traditional water

spectral indices such as the Normalized Difference Water Index (NDWI) [22] and the Modified Normalized Difference Water Index (MNDWI) [23] are commonly used in mapping floods areas and they can be applied effectively. Another index is the recently introduced Flood Mud Index (FMI), which enhances the detection of sediment-laden floodwaters, as demonstrated in the October 2024 Valencia flood event [24]. The integration of remote sensing with Geographic Information Systems (GIS) facilitates flood susceptibility mapping and improves the accuracy of hydrological models, offering a robust framework for urban planning and disaster resilience [25]. These advancements not only enhance post-disaster recovery efforts but also contribute to long-term flood mitigation strategies, reducing the socio-economic impact of extreme weather events in vulnerable metropolitan areas.

Several recent studies have applied Sentinel-1, Sentinel-2, and Landsat 8 data to flood mapping using diverse methodological approaches. Zhang et al. (2020) utilized Sentinel-1 Ground Range Detected (GRD) data acquired in Interferometric Wide (IW) swath mode using (VV) polarization, aimed at rapidly delineating flood extent in semi-arid regions of Pakistan [26]. Peng et al. (2025) proposed a replicable and scalable method implemented on the Google Earth Engine platform that combines Sentinel-1 VH polarization data with orthorectified Landsat 8 surface reflectance imagery and applies NDFVI-based change detection alongside the Edge Otsu thresholding algorithm [27]. Their validation used Sentinel-2 imagery and the Global Surface Water Dataset across the Poyang and East Dongting Lake regions [27]. Similarly, Gašparović et al. (2021) integrated Sentinel-1 (VV and VH polarizations) and Sentinel-2 bands (B2, B3, B4, and B8) to monitor flood events in lowland forest areas, designing a methodology that balances accuracy, processing time, and operational cost [28]. These examples highlight the increasing diversity and refinement of multi-sensor satellite approaches for flood assessment, offering adaptable frameworks across different environmental and geographic contexts.

This study contributes to addressing an identified gap in the flood risk literature concerning the integrated assessment of flood impacts in large Mediterranean metropolitan areas. Although remote sensing techniques are widely used for flood mapping, the integration of multi-sensor satellite imagery with detailed socio-economic and infrastructure exposure data at the metropolitan scale has not yet been widely implemented in scientific and operational frameworks. This study combines multi-sensor satellite data (Sentinel-1, Sentinel-2, and Landsat 8) with socio-economic, land use, and infrastructure datasets to map and analyze the extent and consequences of the October 2024 flood in the Valencia Metropolitan Area. This spatially explicit approach enables the assessment of flood extent, affected population, infrastructure disruption, and land cover impacts. The resulting framework offers a transferable contribution to strengthening urban flood resilience and disaster risk management in Mediterranean settings increasingly exposed to extreme hydrometeorological events.

1.1. Study Area

The Valencia region is situated in the eastern and southeastern part of the Iberian Peninsula (Figure 1), covering an area of 23,255 km^2, which represents about 4.6% of Spain's total surface area and comprises 534 municipalities. It consists of three provinces: Castellon, Valencia, and Alicante. The climate is primarily arid to semi-arid, covering 51% of the territory, with hot, dry summers and rainy autumns. However, in the northern areas, the climate tends to be sub-humid to dry–subhumid, featuring rainy autumns, warm summers, and a notable influence from the Mediterranean Sea [29]. The study's subject is the Valencia Metropolitan Area, which suffered from significant flooding in October 2024 due to intense rainfall storms. It is a densely urbanized zone located on Spain's eastern coast, adjacent to the Mediterranean Sea. It includes the city of Valencia (39°28′0.0012″N, 0°22′30.0000″W) as

its center, together with a network of nearby municipalities that create a cohesive economic and social area.

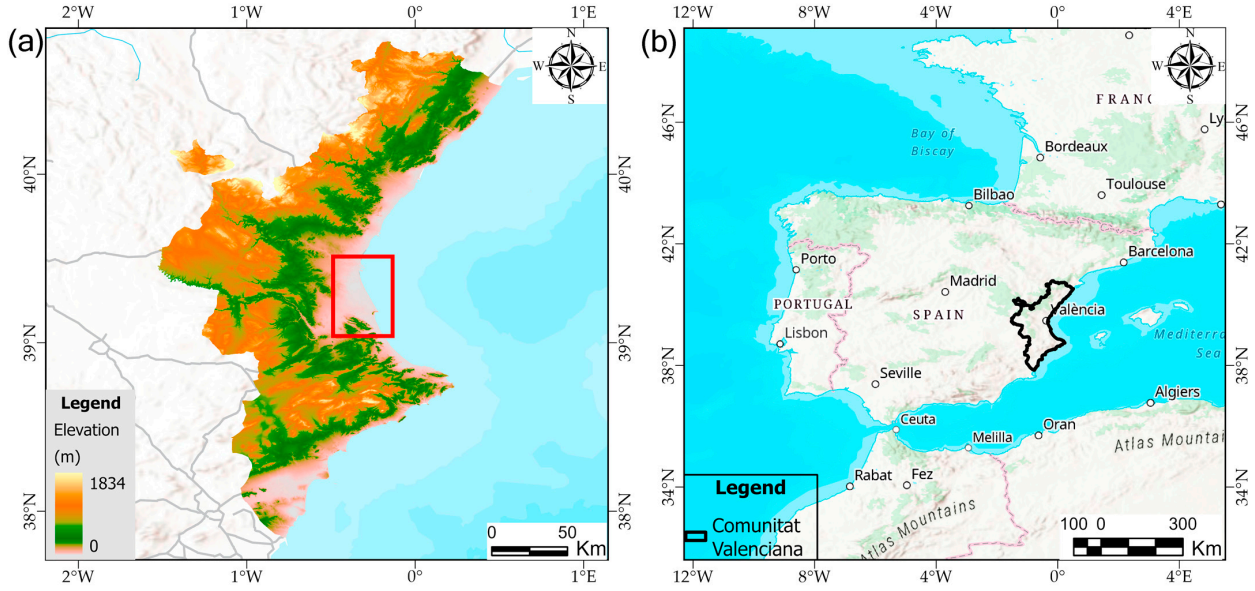

Figure 1. Study area. (**a**) Elevation map of the eastern Iberian Peninsula highlighting the study area (red box) within the southern part of the Metropolitan Area of Valencia. (**b**) Location of the Comunitat Valenciana within the Iberian Peninsula, outlined in black.

The region extends along the Mediterranean coastline, bordered by the Gulf of Valencia to the east. Concerning the administrative division of Valencia, the city consists of 19 administrative regions and 88 administrative communities. The estimated population of Valencia in 2024 was 2,710,808. Valencia has a total area of 10,806 km^2 and a population density of 250.9 people/km^2 [30]. Eastern regions of Spain experience relatively low annual rainfall, with Valencia receiving just over 500 mm annually [13]. The river systems in the Mediterranean region are the most dangerous in Spain due to several factors, including the small size and short length of the river basins, steep terrain, closeness to the coast, and a climate characterized by sporadic rainfall and a tendency for heavy downpours. Historical records show that Valencia has faced numerous flooding events throughout its history, many of which caused substantial damage and loss of life. The River Turia, which ran through the city of Valencia between the Poyo and Carraixet streams, caused a devastating flood in 1957, leading to its diversion from the city [12]. In 1957, a flash flood severely impacted the city center, sweeping away numerous buildings, including historic structures, this event led to an unexpected failure of the metropolitan infrastructure, causing a significant loss of life (81 fatalities, as per official reports) and property damages [31]. That resulted in a housing crisis and made transportation challenging, while also causing the loss of several parks [32]. Another flood occurred in October 1982, affecting the Valencia and Murcia regions. It resulted in 40 deaths and caused damages estimated at EUR 300 million [33].

1.2. The Event of 29 October 2024

On 29 October 2024, the Valencia region experienced a catastrophic meteorological event characterized by an intense cut-off low system, locally referred to as a Depresión Aislada en Niveles Altos (DANA). This cut-off low system led to an intense and localized flooding episode that is considered one of the most impactful in recent decades in the Valencia Region. Its impact is comparable to historically significant flood events in the region, such as those recorded in 1957 and 1982, although a detailed hydrometeorological

comparison is beyond the scope of this study. The Spanish Meteorological Agency (AEMET) had issued an orange weather warning for southern Valencia at 06:42, which escalated to a red warning by 07:36 as conditions deteriorated rapidly. The town of Chiva was among the hardest struck, receiving nearly 500 mm of rainfall within a few hours, overwhelming local drainage systems and causing extensive flooding. Similarly, Turís recorded a historic peak rainfall intensity of 184.6 L per square meter in just one hour, surpassing previous national records [34].

The heavy rainfall led several rivers and ravines, notably the Magro River and the Poyo Ravine, to overflow, inundating multiple municipalities. In Utiel, water levels rose by three meters, trapping residents and causing significant structural damage. The floods disrupted essential services, including electricity and telecommunications, further hampering emergency response efforts. In the aftermath, the floods claimed at least 231 lives, marking one of the deadliest natural disasters in Spain's modern history [35].

Specifically, the floods caused extensive damage to critical infrastructure such as roads, railways, bridges, and public buildings, with the estimated repair costs reaching EUR 2.6 billion [36]. Preliminary assessments by the Valencian Association of Farmers indicate that the agricultural sector in Valencia has suffered losses surpassing EUR 1.09 billion [37]. A sludge action plan worth EUR 500 million will be funded in order to clear accumulated debris and fix water systems in impacted communities [38]. The severity of the event has been linked to the effects of climate change, which are related to exacerbating the frequency and intensity of such extreme weather phenomena [39].

2. Materials and Methods

2.1. Data and Software

The Sentinel-1 mission of the ESA Copernicus Programme is a mission that has collected SAR images around the globe since 2014. The Sentinel-1 satellites are equipped with the C-SAR instrument that operates in the C-Band (5.4 GHz) of microwave area with dual polarization. Sentinel-1 satellites provide images every 6 days, but at the time of the flood, only Sentinel-1A was operational, thus having a 12-day interval. The utilized products were the processed Ground Range Detected (GRD) Level-1 data, acquired in the Interferometric Wide-Swath Mode (IW) and containing amplitude information. The GRD product is focused, processed with multiple looks, projected to the ground range, and includes all bursts and sub-swaths merged. It is also resampled from a 20 m $\times$ 22 m spatial resolution to a uniform pixel spacing of 10 m $\times$ 10 m [40]. The Sentinel-1 images were obtained freely from the Copernicus Data Space Ecosystem (CDSE) platform [41].

Multispectral satellite images from the ESA's Copernicus Programme Sentinel-2 mission were used. The Sentinel-2 mission acquires images every 5 days over a region. The Sentinel-2 satellite is equipped with the MultiSpectral Instrument (MSI), designed to capture reflected solar radiation across 13 spectral bands, spanning from the visible spectrum to the shortwave infrared (SWIR) region [42], with resolutions of 10, 20, and 60 m; 10 and 20 m resolution data were used in this study. The Level-2A products were used, in which an atmospheric correction was applied to obtain the Bottom-of-Atmosphere (BOA) reflectance while also providing scene classification. The image selection relied on the existence of clouds over the area of interest. These images were retrieved from the CDSE platform as were the ones from Sentinel-1 [41,43].

Landsat 8 multispectral images from the NASA/USGS Landsat program were utilized. This satellite, which was launched in 2013, carries the Operational Land Image (OLI) sensor and the Thermal InfraRed Sensor (TIRS) and has a revisit frequency of 16 days. The OLI sensor's enhanced 12-bit radiometric resolution enables it to detect finer levels of detail per pixel than any earlier Landsat sensor. This higher level of quantization makes it possible

to observe subtle changes in surface characteristics [44]. The Landsat 8 images consist of 11 spectral bands, with the bands utilized in this study having a 30 m resolution. More specifically, the products used are from Collection 2 Level-2 processed science products, which are atmospherically corrected and provide surface reflectance information. They are also from Tier 1, which means they are corrected, meeting the highest accuracy standards. The images were obtained freely from the USGS Earth Explorer platform [45].

The CORINE Land Cover 2018 (CLC 2018) land cover and land use dataset was utilized to assess the flood impact on the study area. This comprehensive dataset has a satisfactory accuracy of $\geq$85 % with a Mapping Minimum Width (MMW) of 100 m and a Minimum Mapping Unit (MMU) of 25 ha. It has three hierarchical levels of detail, with the most analytical having 44 land cover classes. The CLC dataset is freely available, and it was retrieved from the Copernicus Land Monitoring Service (CLMS) in vector polygon format [46,47].

The openly available vector datasets of OpenStreetMap (OSM) were used to assess the flood impact on the infrastructure of the study area. The OSM database provides global geospatial datasets created by community contributions. This study used the Geofabrik download server to obtain the OSM data [48,49].

The EU-Hydro dataset was used to map the hydrographic network of the area of study and, more specifically the EU-Hydro River Network Database 2006–2012. This dataset consists of vectors of rivers and streams, waterbodies, and other hydrological data. It was produced predominantly using satellite imagery; it has a 1 ha MMU and is available from the CLMS [50,51].

Regarding population density, the open dataset of WorldPop Population Density 2020 for Spain was used. This dataset is a raster that indicates people per cell, with an approximate 1 km $\times$ 1 km (30 Arc Sec) cell resolution. It is produced with contemporary modeling techniques combining various datasets, e.g., census data, population estimates, and surface area [52,53].

The Digital Elevation Model (DEM) SRTM—Shuttle Radar Topography Mission 1 Arc-Second Global v003 dataset has 30 m pixel resolution with global coverage and corrected for voids (void-filled). It is an openly available dataset, and it was retrieved from the USGS Earth Explorer platform. The ESA STEP SNAP software was used in interpreting the terrain of the study area and in the terrain correction for the SAR images [45,54,55].

For this study, both open and commercial software were utilized. The open remote sensing software, ESA's STEP SNAP v.10–11, was used for the satellite image processing and analysis, while the ESRI's ArcGIS Pro v.3.3–3.4 commercial GIS software was utilized for the processing and analysis of geospatial datasets, results production, and map making.

Tables 1 and 2 list the datasets and satellite images that were used in the study.

Table 1. Datasets used.

Name	Format	Resolution	Source
Sentinel-1	SAR GRD IW	10 m	Copernicus Data Space Ecosystem
Sentinel-2	Multispectral L2A	10–20 m	Copernicus Data Space Ecosystem
Landsat 8	Multispectral C2 T1 L2	30 m	Earth Explorer
SRTM 1 Arc-Sec v3 DEM	Raster	30 m	Earth Explorer
CORINE Land Cover 2018	Vector (Polygon)	-	Copernicus Land Monitoring Service
WorldPop Population Density	Raster	~1 km	Humanitarian Data Exchange
EU-Hydro	Vector (Lines, Polygons)	-	Copernicus Land Monitoring Service
OpenStreetMap	Vector (Lines, Polygons, Points)	-	OpenStreetMap Geofrabrik

Table 2. Utilized satellite images.

Satellite Images	Date	Use
Landsat 8 C2 L2 T	11 August 2024 10:36 UTC	Pre-event
Sentinel-2 L2A	17 August 2024 10:50 UTC	Pre-event
Sentinel-1 GRD IW Ascending	19 October 2024 18:03 UTC	Pre-event
Landsat 8 C2 L2 T1	30 October 2024 10:37 UTC	Post-event
Sentinel-2 L2A	31 October 2024 10:51 UTC	Post-event
Sentinel-1 GRD IW Ascending	31 October 2024 18:03 UTC	Post-event
Sentinel-2 L2A	5 November 2024 10:52 UTC	Post-event

2.2. Satellite Image Processing

In Figure 2, the methodological steps are briefly presented. The steps for the ESA's STEP SNAP v.10–11 are presented in green, and the steps for ESRI's ArcGIS Pro v.3.3–3.4 are presented in cyan.

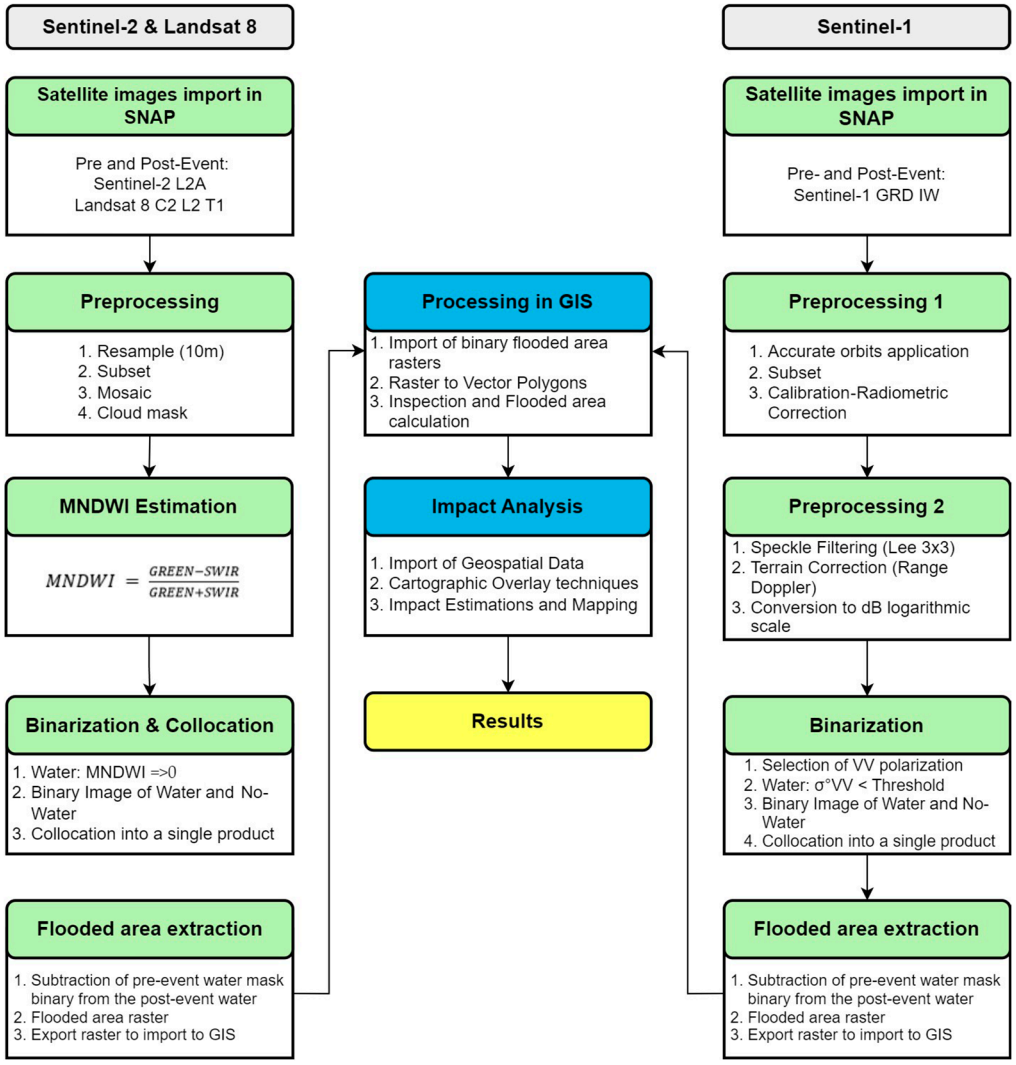

Figure 2. Flowchart of methodology.

2.2.1. Sentinel-1

For flood mapping using Sentinel-1, a sequence of preprocessing steps was performed. SAR data from Sentinel-1 provide a significant advantage over optical sensors, as it can

acquire images in all weather conditions, including during heavy cloud cover, which is common in flood events. The processing workflow involved several key steps, including orbit correction, radiometric calibration, speckle filtering, and terrain correction. These operations ensured that the SAR images were properly aligned, radiometrically consistent, and suitable for further flood extent mapping, enhancing the accuracy and reproducibility of the analysis [56].

The first step in the preprocessing workflow was the application of precise orbit files to improve the geometric accuracy of the images. This correction minimizes geolocation errors and ensures spatial consistency across multiple acquisitions. Following this, a spatial subset of the images was performed, restricting the analysis to the South Valencia Metropolitan Area to optimize computational efficiency. Once the area of interest was defined, radiometric correction (calibration) was applied, using the calibration to σ0 (sigma naught) to enable meaningful quantitative comparisons between different acquisitions.

To minimize the noise present in SAR data, a speckle filtering process was implemented. A Lee speckle filter with a 3×3 window size was applied, selected for its effectiveness in reducing high-frequency noise while preserving edges and flood-related features. After speckle filtering, terrain correction (range Doppler) was conducted using the SRTM 1 Arc-Second Global DEM [55]. The range Doppler terrain correction corrected geometric distortions caused by topography and ensured proper georeferencing to a UTM coordinate system (WGS84, Zone 30N), aligning the SAR data with other geospatial layers [56].

Flooded areas were identified using a thresholding approach based on the backscatter characteristics of water surfaces. The analysis used the Vertical–Vertical (VV) polarization channel, and the σ°VV backscatter values were converted to the decibel (dB) scale. A threshold value was determined based on histogram analysis to select the value with the least frequency between the water and land distributions that separate them [57], and visual interpretation, where areas with lower backscatter values characteristic of open water surfaces were classified as flooded zones. The resulting flood extent map was exported in binary raster format, where flooded areas were represented with a value of 1 and non-flooded areas with a value of 0. This raster was subsequently converted into vector format for integration into the GIS software (ArcGIS Pro v.3.3–3.4), where also utilizing the DEM, a slope map was generated to mask the high slope areas to enhance the results' accuracy.

2.2.2. Sentinel-2 and Landsat 8

For the flood area mapping with the Sentinel-2 and Landsat 8 images, the spectral index MNDWI [23] was applied. This water spectral index is a modification of the NDWI that replaces the Near Infrared (NIR) spectral band with the Short-Wavelength Infrared (SWIR) one and uses the Green (Visible Green) spectral bands (Equation (1)). This index allows water extraction due to the use of Green, which maximizes the typical reflectance of water, and SWIR offers high typical reflectance on land and vegetation, low on water, and enhanced contrast for better classification, reducing misclassifications of land and artificial features. The normalized index has values that range from -1 to 1, with the positive values (=>0) corresponding to water. For the application of MNDWI, the B3 and B11 spectral bands were used in Sentinel-2 images, while for the Landsat 8 images, the B3 and B6 were used [22,23].

$$MNDWI = \frac{GREEN - SWIR}{GREEN + SWIR} \quad (1)$$

In the ESA STEP SNAP v.11, the pre- and post-event multispectral images were imported and then initially resampled into 10 m resolution with the nearest neighbors method for Sentinel-2 and the bicubic method for Landsat 8. The images corresponding to the same date were mosaicked into one, and then a subset was performed to the extent of the Area of Interest (AOI). Only the necessary spectral bands and masks were retained

during this procedure to reduce the processing volume. Then, a cloud mask for each date was created using the respective cloud masks included in the product according to their type and confidence-probability (e.g., cloud shadows, cirrus, clouds, etc.) for each date via application band math. This was then subtracted from the respective image. The next step was to calculate the MNDWI for the pre- and post-event images with band math. Afterward, with the produced MNDWI, the threshold of MNDWI => 0 was applied to create a binary image of 1 corresponding to water and 0 to non-water areas. After the binarization, pre- and post-event products were collocated into one. Then, the pre-event water mask was subtracted from the post-event one to remove the permanent water from the AOI and retain only the floodwater. The final product is a binary raster clear of most clouds and permanent water, and the value 1 represents the flood extent [58].

The binary raster was imported to the GIS software and then converted to vector polygon format. Validation based on the interpretation of both the natural and false-color infrared multispectral images followed. Falsely classified areas were removed from the results before estimating the flooded area and overlaying with the geospatial data [58].

2.3. Impact Analysis with Geospatial Data

Using geospatial data, the flood impact was analyzed using the respective techniques of cartographic overlay in the GIS software. The flood extent derived from the Landsat 8 image was selected for analysis after the comparison with the results derived from the other satellite images, proving that it gives the most representative result when accounting for image availability, date of acquisition, and cloud cover.

The vector polygon results of the flooded area were overlaid with various geospatial datasets for the damage assessment. The linear infrastructure from OSM, which includes the road and rail network, was used to estimate the length affected by the flood. Also, the affected critical infrastructure, including airports, health units, and schools, was pinpointed with the overlay of a 200 m buffer zone around these points with the flooded area extent. The flooded land cover was also estimated using the CLC 2018 dataset [46,47]. Utilizing the WorldPop 2020 population density raster [52], the affected population was estimated. Lastly, a slope map was generated, with rivers and inland water bodies overlaid to improve the visualization of hydrological features in connection with terrain characteristics.

3. Results

3.1. Flood Extent

The flood extent resulting from the October 2024 cut-off low event was mapped using a combination of Sentinel-1, Sentinel-2, and Landsat 8 imagery. The analysis revealed significant inundation across the southern part of the Valencia Metropolitan Area, with a total affected area estimated at approximately 199.03 km^2 based on the 30 October 2024 Landsat 8 image, as presented in Table 3 and Figure 3. The spatial distribution of the flood is characterized by widespread water coverage, primarily concentrated in the flat, low-lying coastal, and peri-urban regions.

Table 3. Estimated flood extent per image acquisition.

Satellite Images	Date	Area (km^2)	Result
Landsat 8 C2 L2 T1	30 October 2024 10:37 UTC	199.03	Analysis Flood Extent
Sentinel-2 L2A	31 October 2024 10:51 UTC	84.16	Flood Extent 2
Sentinel-1 GRD IW Ascending	31 October 2024 18:03 UTC	87.53	Flood Extent 3
Sentinel-2 L2A	5 November 2024 10:52 UTC	Not estimated due to cloud coverage	

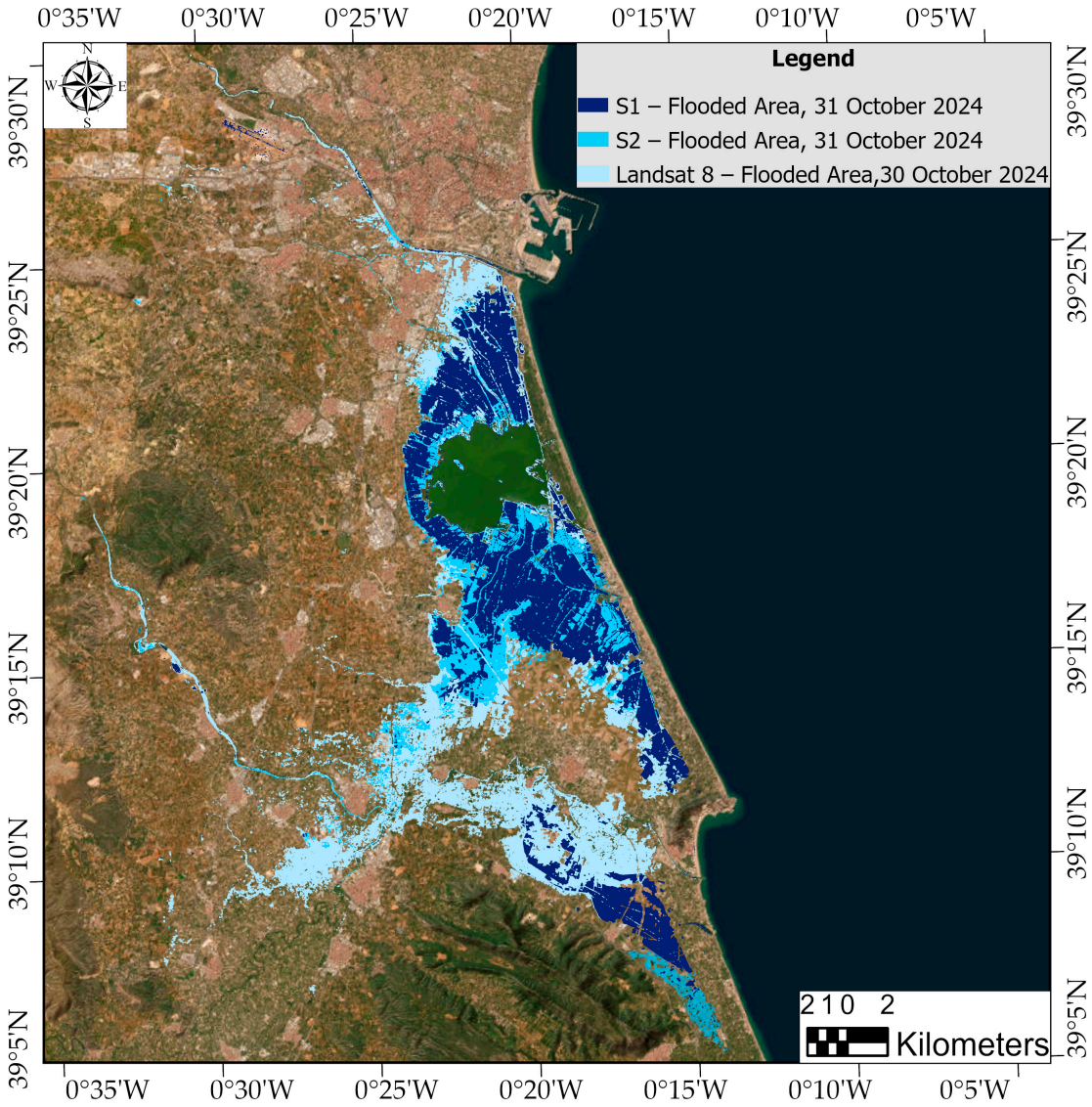

Figure 3. Flood extent in the southern sector of the Valencia Metropolitan Area derived from multi-sensor satellite imagery. The map illustrates the identified flooded areas using Sentinel-1 SAR data acquired on 31 October 2024 (light blue), Sentinel-2 optical imagery from 31 October 2024 (cyan), and Landsat 8 optical imagery from 30 October 2024 (dark blue).

A comparative evaluation of the results obtained from the different satellite sensors highlighted distinct advantages and limitations, as illustrated in Table 3 and Figures 3 and 4. The Landsat 8 extent of 30 October 2024 provided clearer observations with minimal cloud interference, particularly on 30 October 2024, which closely coincided with peak inundation conditions. However, in some areas, due to small and cirrus clouds and their shadows, there was a loss of detail and corrections, as presented in Figure 4a. Next, the subsequent Sentinel-2 acquisition on 31 October 2024 morning, offering high spatial resolution and multispectral bands, resulted in a flood extent of 84.16 km^2, revealing a drainage of −57.71%. Considering the image's characteristics, Figure 4b, the actual drainage is slight, and the result is heavily affected by the prevailing cloud cover, cloud shadows, and cirrus, which were corrected and removed using the related masks. Lastly, the Sentinel-1 SAR imagery on the afternoon of 31 October 2024 (Figure 4c) provided timely information immediately after the flood event, benefiting from its capability to penetrate cloud cover and operate under all weather conditions. Due to its capabilities, the resulting 87.53 km^2 flooded area was 4.00% more than the morning's Sentinel-2 results. However, the SAR-derived flood maps

occasionally exhibited noise, particularly in areas with urban infrastructure and rough surface textures, which could interfere with accurate water detection, as in the study's case.

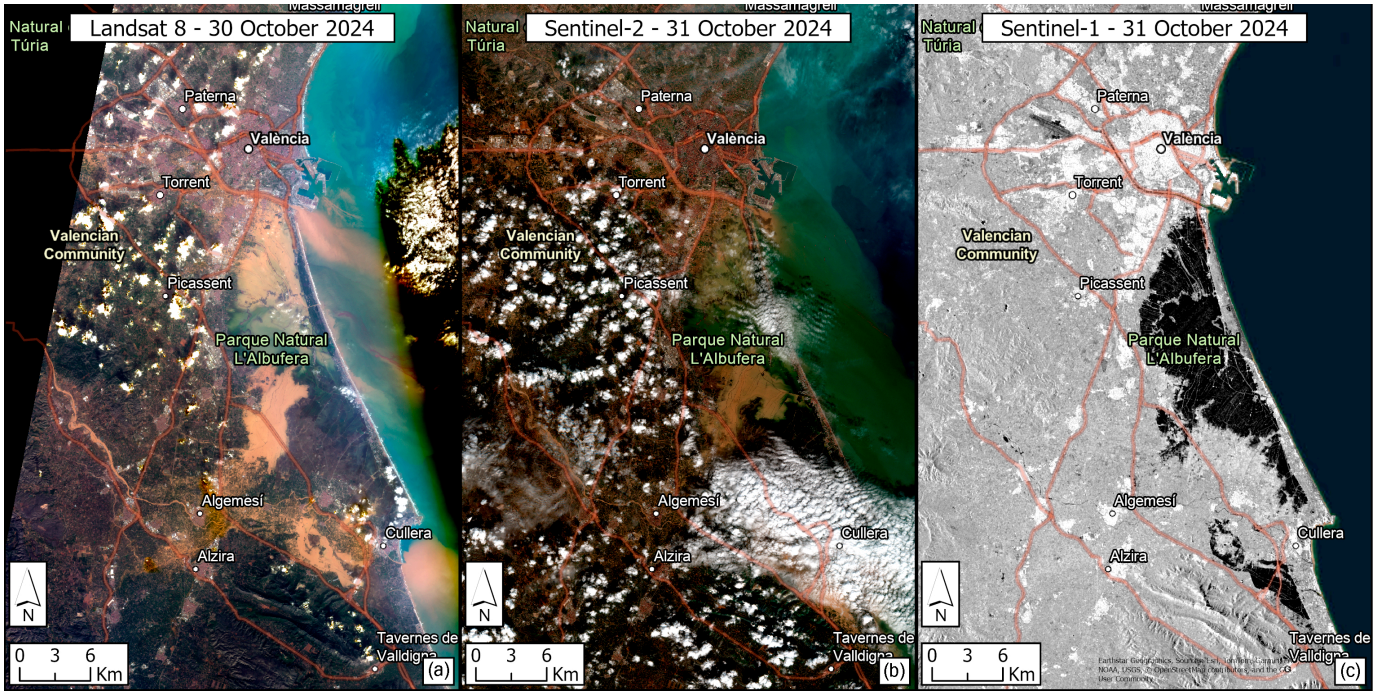

Figure 4. Satellite images used in flood extent delineation. (**a**) Landsat 8—30 October 2024 optical image in natural colors; (**b**) Sentinel-2—31 October 2024 optical image in natural colors; (**c**) Sentinel-1—31 October 2024 SAR image in σ°VV backscatter grayscale.

Considering the Sentinel-2 and Sentinel-1 results, there was actual drainage from within the AOI on 31 October 2024 of approximately 50% of the initial flood extent derived from the Landsat 8 imagery.

Given the favorable atmospheric conditions and temporal proximity to the maximum flood stage relying on the abovementioned analysis, the Landsat 8-derived flood map was selected as the primary reference for subsequent impact analyses. The delineated flood extent from the Landsat 8 imagery clearly illustrates extensive flooding along the coastal plain, the riverine corridors of the Júcar and Magro rivers, and significant portions of the Albufera Natural Park area. Notably, floodwaters encroached into urban municipalities such as Silla, Sollana, and Albalat de la Ribera, underscoring the high vulnerability of the peri-urban fringe zones.

3.2. Flood Impact Analysis

A detailed population exposure assessment was conducted by overlaying the Landsat 8-derived flood extent with the WorldPop 2020 population density raster [49]. Using zonal statistics, the number of residents located within the inundated areas was estimated to quantify the human impact of the event. The results indicate that approximately 90,000 inhabitants were directly affected by flooding across the southern Valencia Metropolitan Area.

The spatial analysis reveals that areas with moderate to high population densities (100 to >1000 inhabitants/km^2) were particularly impacted. Urban municipalities such as Silla, Sollana, Albalat de la Ribera, and Benicull de Xúquer exhibited the highest levels of exposure, with floodwaters encroaching upon densely populated neighborhoods (Figure 5). Additionally, significant population clusters in the municipalities of Sedaví, Massanassa, and Alfarb were also affected, albeit to a lesser extent.

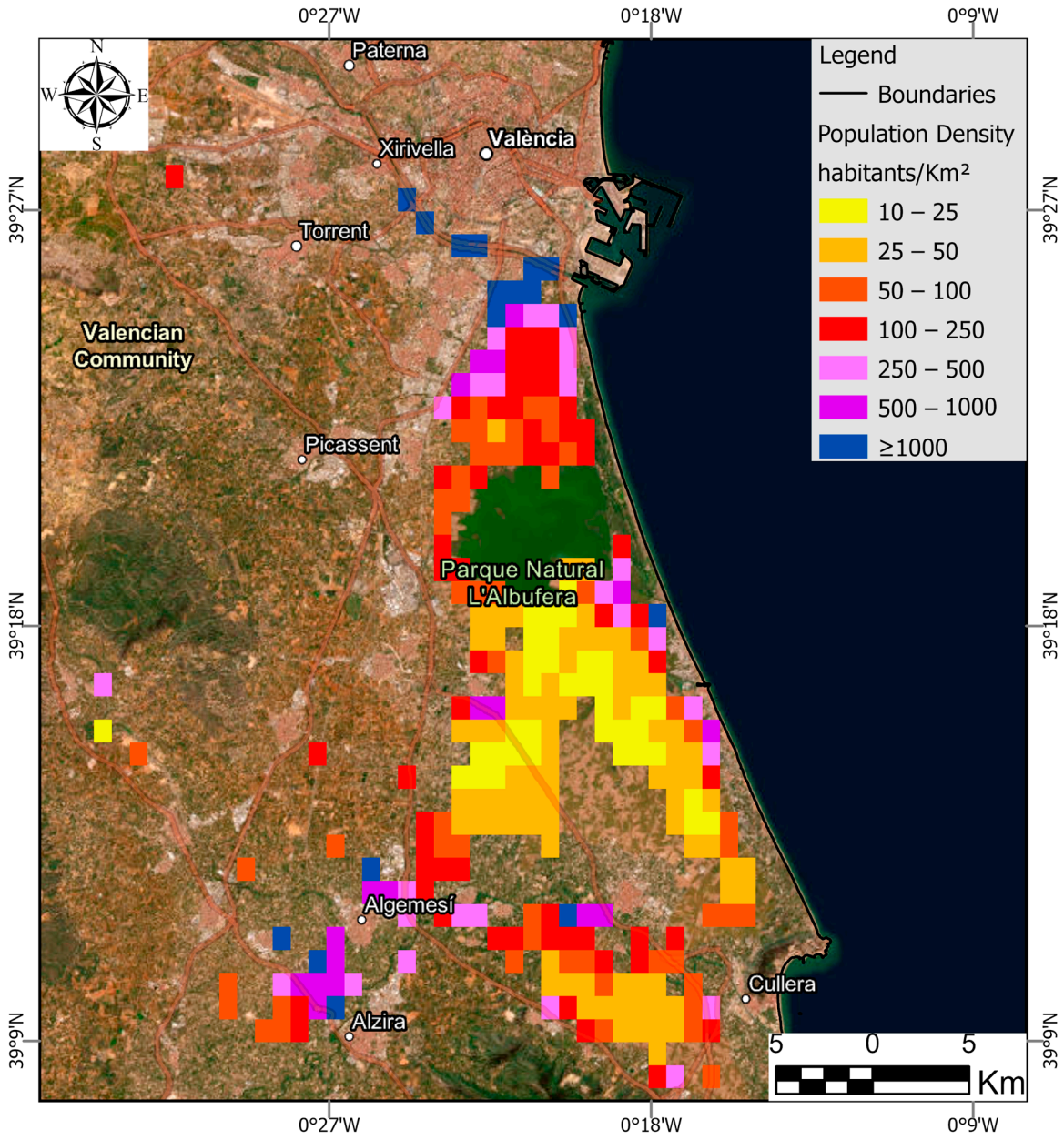

Figure 5. Population density distribution in the Valencia Metropolitan Area based on WorldPop 2020 data [52] overlaid with the 30 October 2024 Landsat 8 flood extent. The map presents the population density values classified into six categories ranging from 10 to over 1000 inhabitants per km^2.

This pattern is consistent with previous studies highlighting the vulnerability of Mediterranean urban environments, where rapid urban expansion often extends into historically flood-prone areas. The concentration of affected populations in peri-urban zones reflects the interplay between geomorphological factors (low slope, proximity to river floodplains) and socio-economic drivers such as urban sprawl and land use planning deficiencies.

In addition to assessing population exposure, the vulnerability of critical infrastructure was evaluated to determine the broader socio-economic impact of the flooding. Key facilities, including health units, hospitals, schools, airports, and transportation networks, were spatially analyzed in relation to the flood extent. A 200-meter buffer zone was established around each infrastructure point to capture direct and indirect exposure.

The analysis revealed that seven hospitals and health centers, primarily located in Valencia, Benetusser, Albal, and Cullera municipalities, were either directly inundated or

within proximity to flooded areas (Figure 6). This proximity increases the likelihood of service disruption due to accessibility issues, damage to utility services, or indirect effects due to nearby road closures.

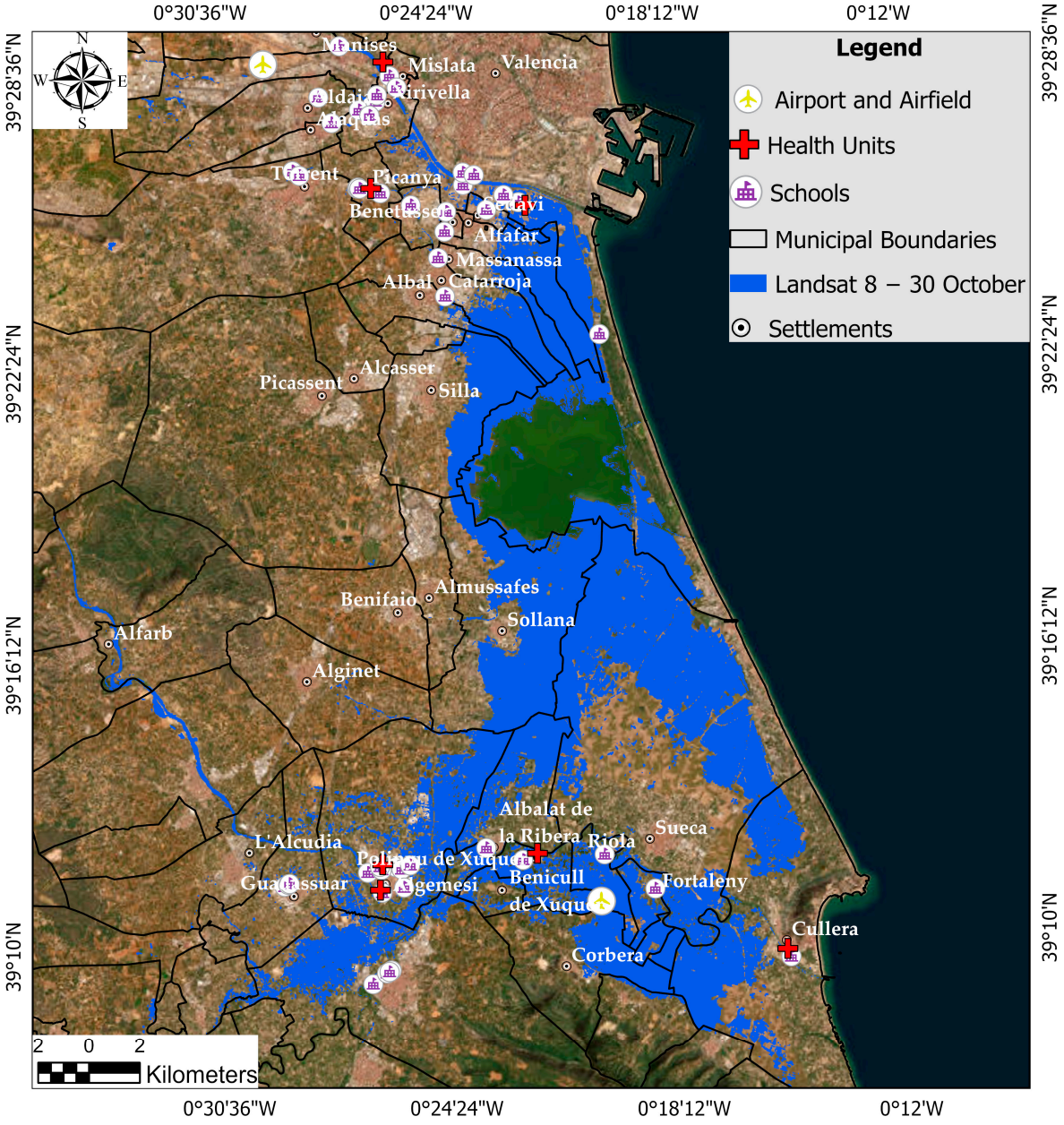

Figure 6. Affected critical infrastructure derived from the overlay of the 30 October 2024 Landsat 8 flood extent with point locations of airports, health units, and schools within the inundated areas or their immediate surrounding zones.

Regarding educational facilities, fifty schools within the Valencia Metropolitan Area were identified as being affected by the floodwaters. Most of these schools are concentrated in moderately to highly populated urban zones, including Sedaví, Silla, Sollana, and Benicull de Xúquer, consistent with general trends in clustering educational infrastructure near population centers.

Furthermore, the analysis identified that two airports, namely Valencia Airport and Alfafar Airfield, are located within areas vulnerable to flooding. While significant flooding was not observed directly over runways, the proximity of floodwaters to access roads and supporting infrastructure could significantly hinder operations.

The spatial intersection of critical infrastructure with flood-affected areas emphasizes the systemic vulnerability of the urban fabric in the Valencia region. These findings highlight the necessity of incorporating flood hazard assessments into infrastructure planning and resilience strategies, especially in Mediterranean coastal regions where such hydrometeorological events are expected to become more frequent.

To further assess the impact of the October 2024 flood event, the delineated flood extent was overlaid with the CORINE Land Cover 2018 dataset to evaluate the affected land use types. The analysis revealed that the flooded areas encompass diverse land cover categories, reflecting both urban and rural land uses in the southern Valencia Metropolitan Area (Table 4).

Table 4. CORINE Land Cover 2018 of flooded areas.

CORINE Land Cover	Percentage %
Continuous urban fabric	0.17
Discontinuous urban fabric	0.26
Industrial or commercial units	1.34
Road and rail networks and associated land	0.00
Port areas	0.08
Airports	0.01
Mineral extraction sites	0.18
Construction sites	0.21
Sport and leisure facilities	0.03
Permanently irrigated land	1.78
Rice fields	64.97
Fruit trees and berry plantations	25.86
Pastures	0.13
Complex cultivation patterns	1.76
Land principally occupied by agriculture with significant areas of natural vegetation	0.19
Coniferous forest	0.09
Natural grasslands	0.36
Sclerophyllous vegetation	0.24
Transitional woodland–shrub	0.00
Beaches, dunes, sands	0.28
Sparsely vegetated areas	0.00
Inland marshes	0.12
Salt marshes	0.60
Water courses	0.63
Water bodies	0.26
Coastal lagoons	0.47
Total	100

One of the most significantly impacted categories corresponds to rice fields, which accounted for approximately 129.17 km^2, representing 64.97% of the total flooded area. This result aligns with the geographical distribution of rice cultivation in the Albufera region.

That area is traditionally dedicated to irrigated agriculture and particularly susceptible to flooding due to its flat topography and proximity to coastal lagoons.

Similarly, fruit trees and berry plantations covered 51.41 km^2 (25.86%) of the inundated area. These agro-productive zones are economically important for the region, and their exposure to floodwaters may result in crop loss, soil degradation, and long-term impacts on agricultural productivity.

In terms of urban areas, continuous urban fabric (dense residential zones) accounted for 0.33 km^2 (0.17%), while discontinuous urban fabric (less dense suburban areas) represented 0.51 km^2 (0.26%). Although their percentage appears relatively low compared to agricultural land, these zones host critical population centers and infrastructure, increasing the socio-economic vulnerability of the area.

Industrial and commercial units were also affected, covering 2.67 km^2 (1.34%) of the flooded extent, particularly around the municipalities of Almusafes and Almácera, home to key manufacturing and logistics facilities. Additionally, construction sites (0.41 km^2), mineral extraction sites (0.36 km^2), and sports and leisure facilities (0.06 km^2) experienced inundation.

Regarding natural landscapes, the floodwaters impacted areas such as natural grasslands (0.72 km^2), sclerophyllous vegetation (0.47 km^2), and salt marshes (1.19 km^2), with potential ecological implications for habitat stability and biodiversity.

Overall, the land cover analysis underscores the multifaceted nature of flood impacts in the region, affecting not only urban areas and infrastructure but also critical agricultural sectors and natural ecosystems (Figure 7).

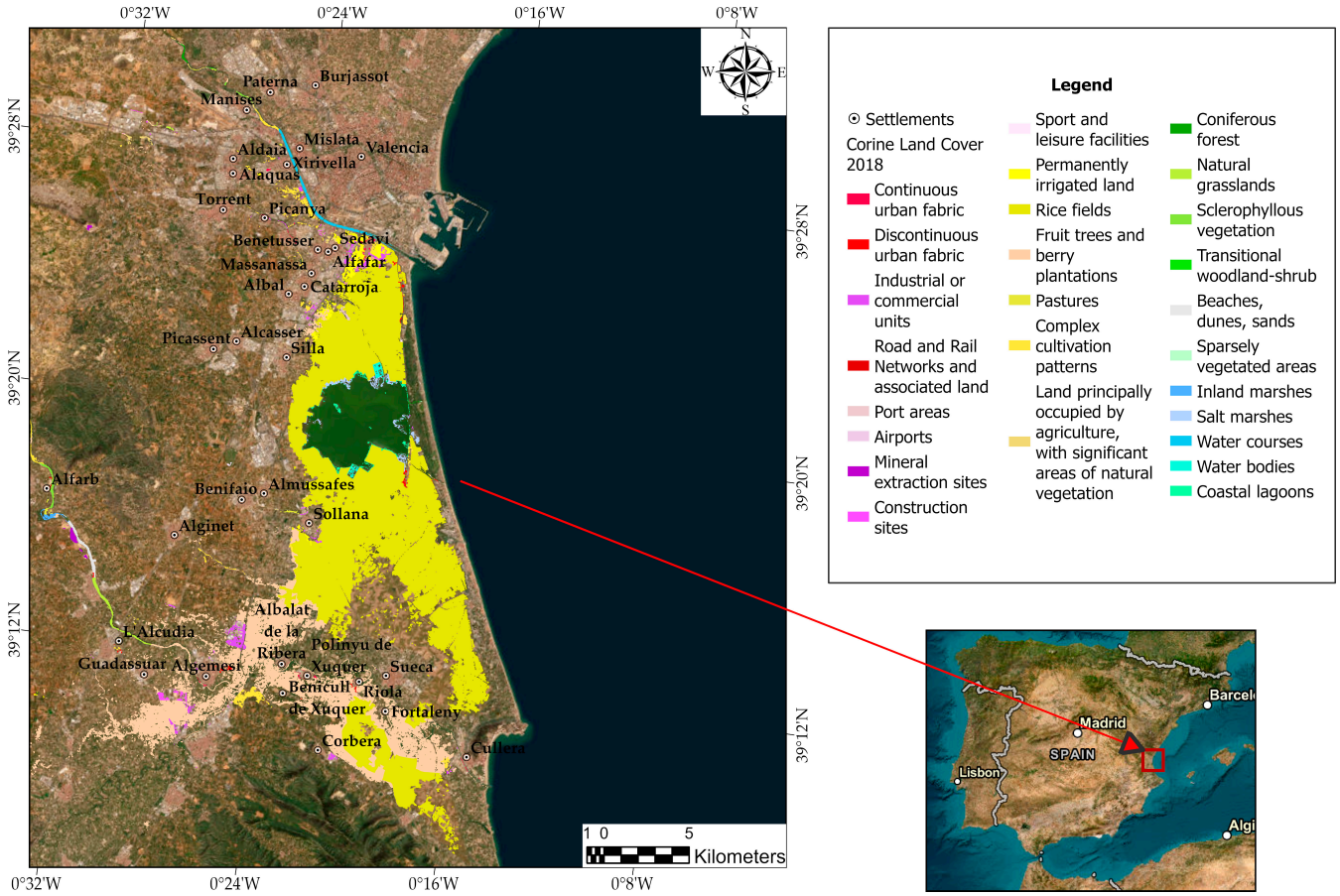

Figure 7. Land use classification of the study area based on the CORINE Land Cover 2018 dataset [43] and the 30 October 2024 Landsat 8 flood extent. The inset map shows the location of the Valencia Metropolitan Area within Spain.

The disruption of transportation infrastructure is one of the most critical consequences of flood events, affecting mobility, emergency response, and supply chains. In this study, the Landsat 8-derived flood extent overlay with OSM vector data allowed for a detailed analysis of the impact on the road and rail networks within the Valencia Metropolitan Area.

The results, illustrated in Figure 8, reveal that approximately 30.53 km^2 of railway lines, 14.24 km^2 of highways and primary roads, and 15.15 km^2 of additional transport-associated land were directly affected by flooding. Major transportation arteries, including sections of the A-7 and AP-7 highways, experienced inundation, particularly in the southern corridors near Sollana, Sueca, and Albalat de la Ribera. These highways represent crucial north–south links in eastern Spain and are essential for transporting goods and people across the region.

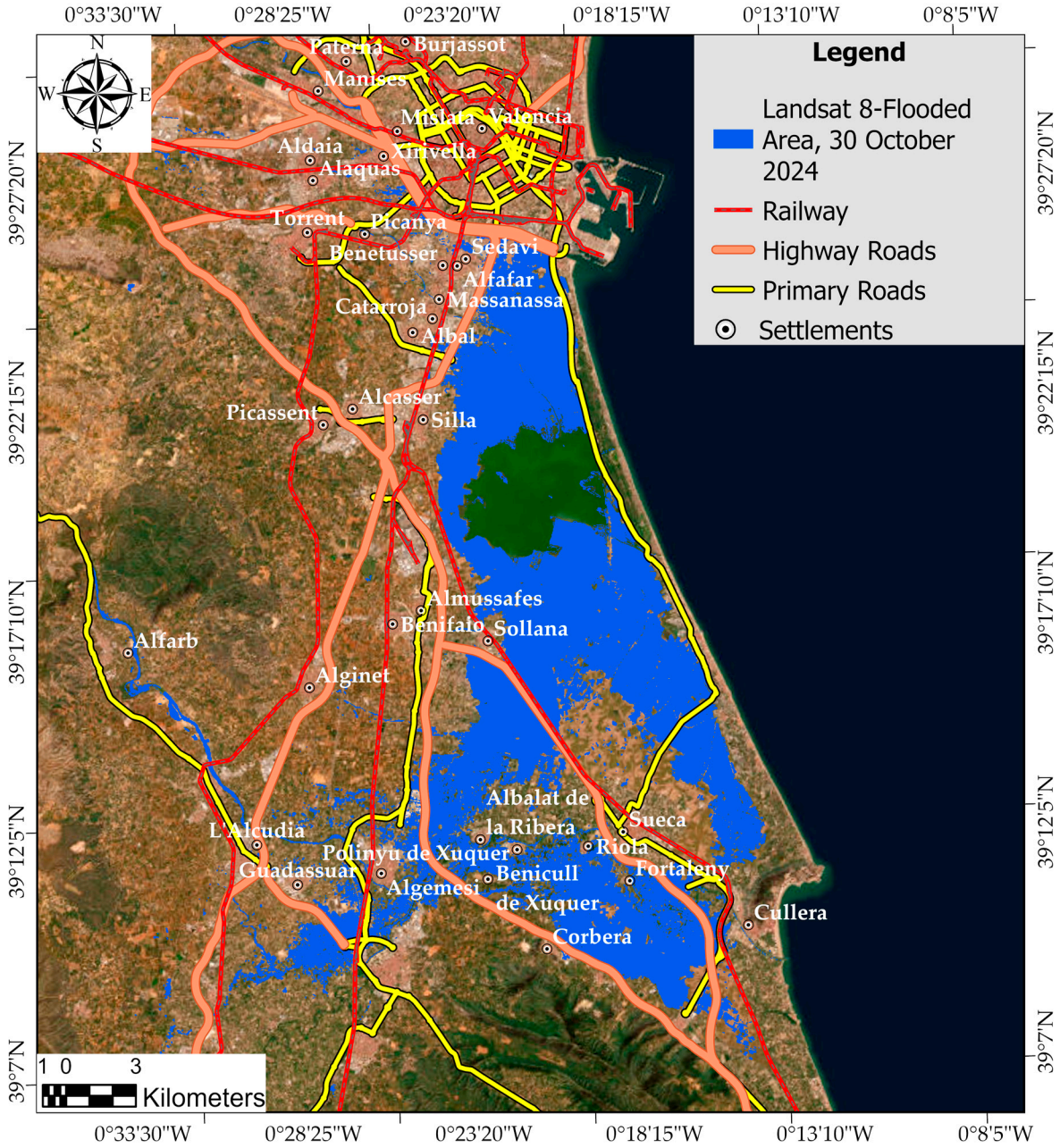

Figure 8. Transportation infrastructure, including railway lines, highways, and primary roads overlaid with the 30 October 2024 Landsat 8 flood extent.

Similarly, significant segments of the rail network, especially those connecting Valencia city with southern municipalities and the port area, were within flooded zones. The

flooding of railway tracks poses not only immediate risks due to track instability and service interruption but also long-term impacts on logistics and economic activity.

Furthermore, secondary roads and municipal streets in peri-urban areas such as Benicull de Xúquer, Polinyà de Xúquer, and Almussafes were also submerged, likely complicating local access and evacuation efforts.

The hydrogeomorphological characteristics of the Valencia Metropolitan Area play a fundamental role in determining the spatial distribution and severity of flooding events. To better understand these dynamics, the delineated flood extent was analyzed in conjunction with terrain slope data derived from the SRTM 1 Arc-Second Global DEM and the hydrographic network sourced from the EU-Hydro dataset.

The slope map (Figure 9) reveals that most flooded areas coincide with zones exhibiting slopes of less than 3%, particularly in the coastal plains south of Valencia, including municipalities such as Sollana, Albalat de la Ribera, Sueca, and Cullera. These flat, low-lying regions are especially prone to water accumulation during periods of intense precipitation, as they lack sufficient natural gradient for rapid drainage.

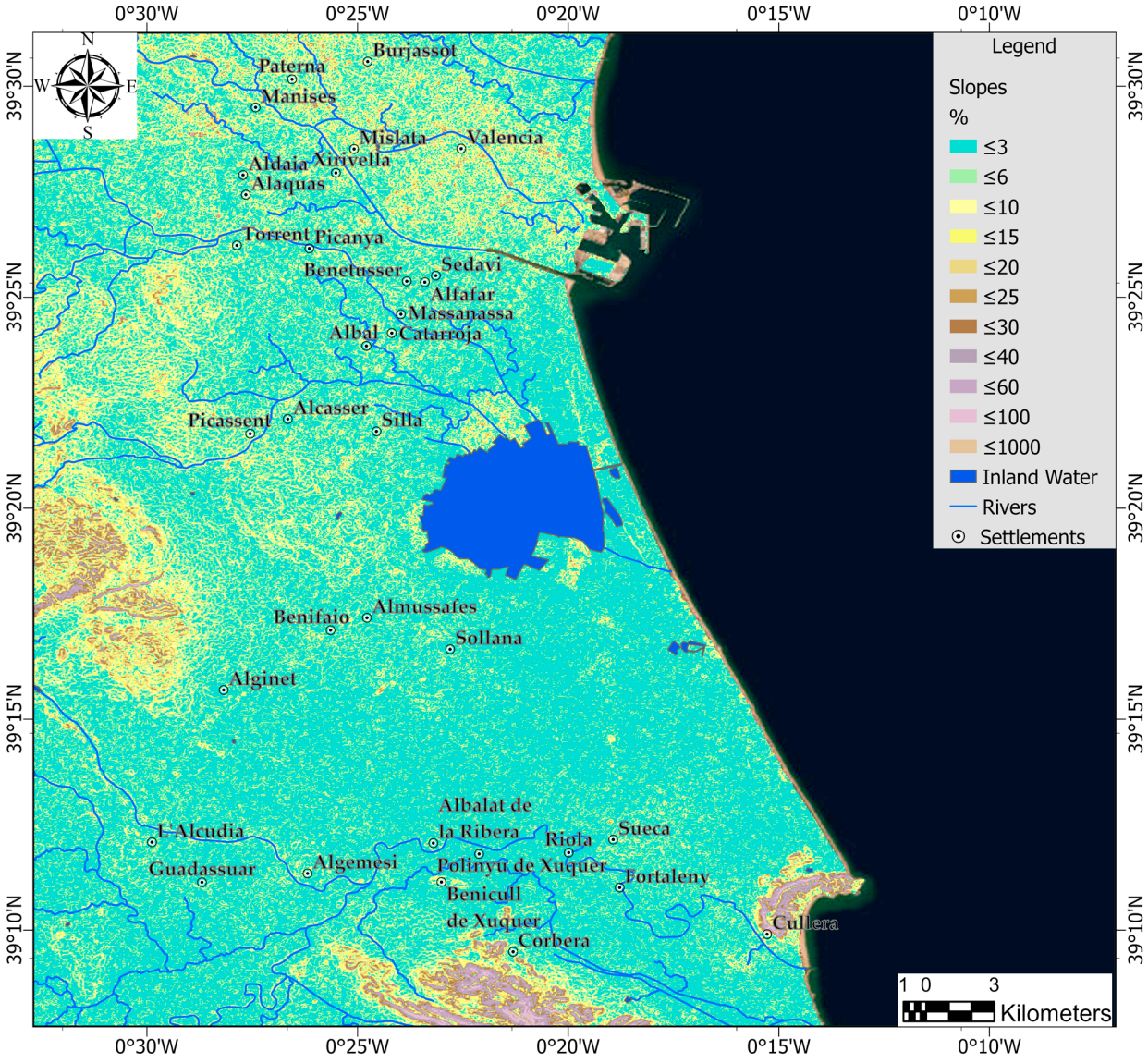

Figure 9. Slope map of the Valencia Metropolitan Area and surrounding regions including its hydrology and settlements. The slope gradients, derived from digital DEM data [43], are presented as percentages (%) and classified into multiple classes.

Additionally, the hydrographic network highlights the role of key rivers, specifically the Júcar River, Magro River, and their tributaries, in shaping the flood patterns. These rivers traverse narrow floodplains and are characterized by small, steep upstream basins and short response times, which, combined with the heavy rainfall associated with the October 2024 cut-off low, contributed to rapid runoff and subsequent overbank flooding.

Of particular note is the Albufera Natural Park, a large coastal lagoon and wetland system, which acted as a natural reservoir for excess floodwaters. However, its surrounding agricultural and urbanized areas, particularly the rice fields, also experienced significant inundation due to the area's flat geomorphology and limited drainage capacity.

3.3. Validation

To validate the satellite-derived flood extent, the delineation product provided by the Copernicus Emergency Management Service (CEMS)—Rapid Mapping EMSR773 was utilized [59]. Specifically, the dataset delivered on 2 November 2024 at 03:02 UTC (Version 1) was employed, offering authoritative and timely geospatial information for benchmarking flood boundaries.

The pre-flood reference image was acquired by Sentinel-2B on 12 August 2024, at 10:46 UTC, featuring a 10 m spatial resolution. For post-flood conditions, images from Sentinel-1A (acquired 31 October 2024, at 18:02 UTC, 20 m resolution) and Sentinel-2B (acquired 31 October 2024, at 10:51 UTC, 10 m resolution) were utilized, with the latter serving as the base map. Additionally, a post-flood Landsat 8 image from 30 October 2024, at 10:37 UTC, with a 30 m resolution was used. Based on the post-event satellite data, the flood extent thematic layer was generated via a semi-automated process. It should be noted that accuracy may be reduced in densely built or forested areas due to limitations inherent in SAR data processing, especially where optical images are unavailable or of poor quality. Information regarding flood water depth was derived by combining the analysis of the latest satellite imagery with the digital elevation model data. The Copernicus product estimated a flooded area of 15,633.4 hectares (equivalent to 156.33 km^2), which aligns closely with the study's estimation of approximately 199 km^2, indicating a reasonably sufficient level of agreement and supporting the reliability of the derived results. Spatially, the CEMS results underestimated the flooded area, such as the area around Sueca, which was characterized by the analysis being classified as a flood trace. These differences are due to different methodological frameworks used, while the results spatially agree in most cases.

To evaluate the accuracy of this study's flood extent estimation, results derived from Landsat 8 imagery were compared with publicly available flood maps. Specifically, data from the SNCZI platform [60] were utilized, which provide official flood hazard and risk assessments, including T = 500 return period flood maps (Figure 10). This map represents areas expected to be inundated during a 500-year recurrence interval flood event and serves as a comparison reference. The validation process involved analyzing the flooded area estimates from this study against those provided by the T = 500 return map. The flood extent derived from Landsat 8 imagery was calculated at ~199 km^2, whereas the T = 500 dataset indicated a significantly larger inundation area of approximately 757.93 km^2. However, the area flooded in October 2024 is within the T = 500 flood extent, thus proving the risk in the area. This notable difference suggests that the flood extent in T = 500 is more severe and widespread, exceeding the area affected by the flood event observed in October 2024.

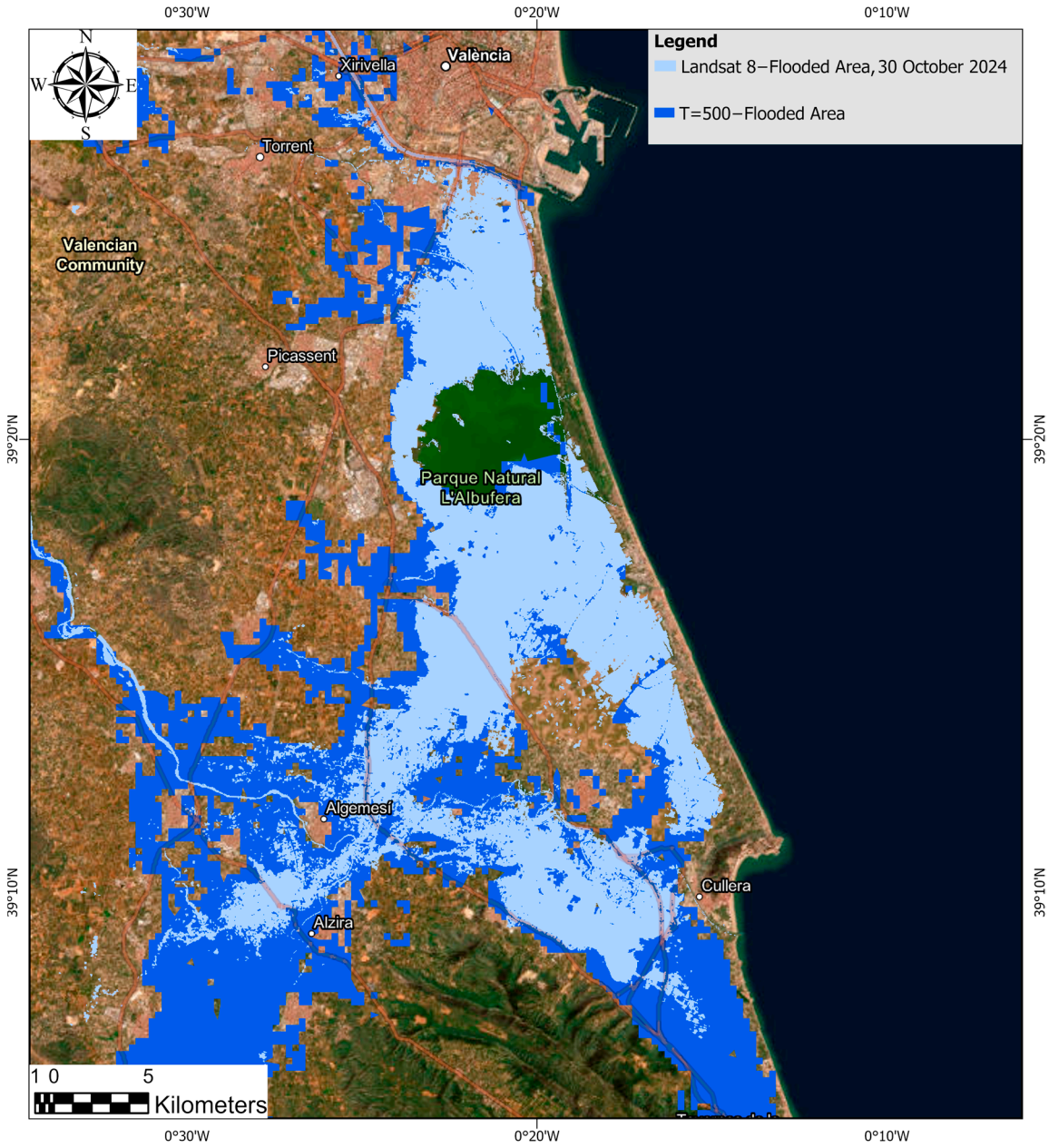

Figure 10. Overlay of flooded areas detected in the Valencian Community (Spain) using Landsat 8 imagery (30 October 2024) and modeled extent from the T = 500 scenario. Light blue indicates satellite-observed flooding, while dark blue represents the T = 500 simulated flooded area.

The increased extent of flooding may be attributed to more intense hydrometeorological conditions, prolonged precipitation, or landscape modifications impacting water retention and drainage patterns. These findings emphasize the critical role of satellite-based remote sensing in monitoring flood events and improving risk assessments. In future studies, flood mapping accuracy could be enhanced by integrating higher-resolution Sentinel-1 SAR data and hydrodynamic modeling techniques to refine the delineation of inundated areas.

4. Discussion

The multi-sensor remote sensing approach employed (Sentinel-1 SAR, Sentinel-2, and Landsat 8 Multispectral) proved effective for flood mapping, with each sensor offering distinct strengths and weaknesses. Sentinel-1 provided all-weather, day–night imaging

immediately during and after the flood, crucial for timely mapping despite pervasive cloud cover. Its C-band SAR detected inundation across the Valencia plains even under cloudy conditions; however, speckle noise and urban backscatter effects introduced some uncertainty. Sentinel-2 allowed a detailed analysis of flood extent and the use of spectral water indices once the skies cleared. It captured finer features better than SAR. However, the limitation of optical data was that the heavy cloud cover during the flooding hindered Sentinel-2 observations, delaying optical-based mapping until post-event clear-sky acquisitions.

Landsat 8 images are those available closest to the flood peak. The resulting imagery provided a clear synoptic view of the extent of inundation at high water levels. That meant the Landsat 8-derived flood map was comprehensive, capturing flooded areas that other sensors might have missed or where flood water had receded. The study selected the Landsat 8 flood extent as the primary reference for impact analyses due to its optimal trade-off between coverage and timing. Comparing across sensors, the flood outlines from Sentinel-1 and Sentinel-2 generally agreed with the Landsat 8 map on the broader inundation pattern, reinforcing the reliability of the mapping. Overall, the integrated use of radar and optical imagery was advantageous; the Sentinel-1 SAR ensured no gap in observation during the clouds, while optical data (Sentinel-2 and Landsat 8) allowed validation and refinement of flood boundaries when weather permitted. That highlights a key insight for flood remote sensing: a multi-sensor strategy can overcome the limitations of any single satellite, improving both the timeliness and accuracy of flood mapping.

Water depth is also a critical factor in flood damage assessment. Karambiri et al. (2015) [61] reported that structural damage begins at a water depth of 0.2 m, with the highest damage rate of approximately 40% occurring at a submersion depth of around 1 m. Also, according to Karagiannis et al. (2017) [62] the extent of damage, including the associated repair costs and time required for restoration, has been shown to rise in correlation with both the depth of floodwaters and the length of time the area remains submerged.

4.1. Socio-Economic Impact Assessment

The October 2024 flood had widespread socio-economic repercussions in the Valencia Metropolitan Area, affecting the population, critical services, agriculture, and transportation. Population exposure was high: an estimated 90,000 residents were within the inundated areas, according to the overlay of the flood map with population data. Notably, the worst-affected communities were peri-urban municipalities south of Valencia (e.g., Silla, Sollana, Albalat de la Ribera, Benicull de Xúquer), where floodwaters encroached into densely populated neighborhoods. These areas correspond to zones of recent urban expansion onto low-lying floodplains, reflecting a pattern observed in many Mediterranean cities where development has sprawled into historically flood-prone lands. The concentration of flood impacts in such peri-urban localities underscores the vulnerability created by unchecked urban growth in floodplains.

The flooding also affected critical infrastructure, amplifying the flood's socio-economic consequences. Spatial analysis revealed that at least seven major healthcare facilities (hospitals and primary health centers) were either directly inundated or within 200 m of flooded areas. These included clinics in the city of Valencia and nearby towns (Benetússer, Albal, Cullera), where flood proximity disrupted services and access. Similarly, educational infrastructure suffered: approximately 50 schools across the metropolitan area were affected in varying degrees by floodwaters. Many of these schools are within the same hard-hit suburbs, meaning thousands of students faced school closures and facility damages. The flood's timing compounded the impact, coming during the academic year and causing

significant disruption to education in those communities. Furthermore, two airports (Valencia's main airport and the smaller Alfafar airfield) were situated at the edge of the inundated zone. The intersection of flood extent with critical infrastructure highlights how even a relatively small percentage of flooded urban land can translate into outsized socio-economic disruption when it affects essential services. The need to close roads and utilities near hospitals and schools, even if those facilities were not completely flooded, illustrates the cascade of indirect impacts (power outages, inaccessible healthcare, halted education) accompanying physical inundation.

The flood's impact on agriculture and rural livelihoods was particularly severe, given that most of the inundation occurred on farmland. Land use analysis showed that expansive agricultural zones were submerged, with rice paddies being the most affected land type—about 129 km^2 of rice fields were flooded, representing ~65% of the total mapped flood extent. Orchards and citrus groves were another significant consequence—roughly 51 km^2 of fruit tree plantations (about 26% of the flooded area) were inundated. Prolonged submergence of orchards can lead to root damage and reduced yields in subsequent seasons, so the flood portends longer-term losses in agricultural productivity beyond the immediate crop destruction. The devastation of these agro-productive zones is socio-economically significant, as agriculture remains an important sector in Valencia's peri-urban fringe. Many rural communities around Valencia rely on rice and horticulture; thus, the flood caused immediate financial damage and jeopardized livelihoods and food supply chains in the region.

Transportation infrastructure suffered extensive disruption, compounding the socio-economic impact by isolating communities and halting commerce. The overlay of flood maps with OpenStreetMap data revealed that large stretches of roads and railways were inundated. In total, a length of~15k m of major highways, ~15 km of primary roads, and ~30 km of rail lines lay within the flood zone. Critical corridors, including sections of the A-7 and AP-7 highways—the primary north–south highways along Spain's east coast—were underwater, especially near Sollana, Sueca, and Albalat de la Ribera. Highway flooding stalled regional traffic and impeded emergency response and relief convoys, which struggled to reach the hardest-hit localities. The rail network was similarly affected: segments of the Valencia–Alicante railway and lines serving the Port of Valencia were flooded. Service was suspended as waterlogged tracks and potential tracked erosion made the passage unsafe. That caused significant cancellations of both passenger and freight trains, with economic ripples in commerce and daily commuting. This multi-sector disruption (population displacement, critical service outage, agricultural loss, and transport shutdown) demonstrates the cascade effects of a major flood in a metropolitan region. Therefore, effective disaster risk management must address not only the hazard extent but also the intertwined vulnerabilities in population centers, infrastructure networks, and economic sectors.

4.2. Comparative Analysis with Other Mediterranean Flood Events

The October 2024 Valencia flood can be contextualized by comparing it to similar extreme flood events in the Mediterranean region in recent years, including cases in Greece, Italy, and France. From a meteorological perspective, the Valencia event was driven by a cut-off low-pressure system (locally known as DANA) that became quasi-stationary over eastern Spain—a classic setup for intense convective rainfall in the western Mediterranean. That resulted in extraordinary precipitation totals (nearly 500 mm in a few hours in some locales) and subsequent flash flooding. A comparable mechanism caused Greece's Thessaly flood in September 2023: Storm Daniel was a slow-moving low that drew energy from anomalously warm sea surfaces, producing an average of ~360 mm of rain over four days

(with local maxima well above 500 mm) [58]. According to satellite assessments, the prolonged heavy rainfall from Daniel led to catastrophic flooding in Thessaly, inundating a vast area—over 1000 km^2 [58]. While Storm Daniel had some characteristics of a *Medicane* (Mediterranean tropical-like cyclone), and the Valencia DANA was a cold-core upper-level low, both systems underscore the Mediterranean's propensity for cut-off lows and quasi-tropical storms that stall and unload enormous quantities of rain.

In Italy, the devastating floods in Emilia–Romagna in May 2023 were triggered by a similarly unusual synoptic pattern: an intense, nearly stationary cyclone migrated from North Africa and lingered over central Italy [63]. Over 2–3 days, it released >300 mm of rain on already-saturated ground, causing 23 rivers to burst their banks simultaneously, causing extensive flooding in that region. This Italian event differed in that antecedent conditions (earlier rains) played a key role—soils could not absorb the new rainfall, greatly amplifying runoff [63]. Furthermore, Mediterranean France regularly experiences intense autumn rainstorms. For example, a cut-off low in October 2018 struck the Aude region of southern France, dropping 244 mm of rain in just 6 h overnight. The result was a series of flash floods that tore through villages, illustrating how quickly such deluges can turn deadly in the rugged terrain of coastal France (at least 15 fatalities were recorded in that event) [64].

In summary, the Valencia flood's meteorological driver—a cut-off low yielding extreme convection—is part of a broader pattern of Mediterranean extreme precipitation events, whether manifested as Spanish DANAs, Greek *medicanes*, or Italian and French autumn storm systems. All share the capacity for short-duration, high-intensity rainfall that overwhelms local waterways. Climate analyses have noted an increasing trend in these high-impact hydrometeorological phenomena in southern Europe, potentially linked to warming conditions due to climate change [2].

Despite these commonalities, the flood extent and urban impacts varied across the events, reflecting differences in geography and exposure. The flooding in Valencia primarily affected a broad low-lying plain (around 200 km^2 inundated, predominantly agricultural flatlands), whereas the Thessaly flood in Greece spread across an even larger agricultural basin with alluvial background (submerging about 1000 km^2 including both eastern and western sub-basins) [15,58]. In contrast, the Emilia–Romagna floods in Italy were distributed across multiple river valleys; though the aggregate flooded area was substantial, it was fragmented, with dozens of towns experiencing localized flooding rather than one contiguous inundation. The Italian event also involved extensive landslides (over 300) due to hilly terrain, compounding damage—a hazard less relevant on Valencia's flat terrain [63]. Additionally, southern France's flash floods (like Aude 2018) tend to be highly concentrated along stream channels and downstream villages; the total flooded area in such cases is relatively small, but the water depth and flow velocity in narrow corridors cause severe destruction to infrastructure in situ.

Urban areas bore heavy consequences in all events, but the scale differed: the Valencia flood encroached into several mid-sized towns on the metro fringe and some peripheral neighborhoods of Valencia city while sparing the dense city center. In Greece, parts of the city of Larissa (pop. ~150,000) were inundated when the Pineios River overflowed, and numerous smaller towns, like Palamas, and settlements were completely submerged for days. Italy saw medium cities like Faenza, Cesena, and Forlì inundated by river floods and tens of thousands of residents evacuated across the region. France's Aude event devastated small towns (e.g., Trèbes, Villegailhenc), with fast-moving waters destroying houses and roads.

The human and economic impacts are also important to compare. The October 2024 Valencia flood stands out as exceptionally deadly—official reports attribute at least

231 fatalities to the event, making it one of the worst natural disasters in modern Spanish history. This tragic toll far exceeds that of the other recent Mediterranean floods: Storm Daniel (Greece 2023) caused 17 deaths in Greece (and additional casualties in Turkey and Bulgaria), while the Emilia–Romagna 2023 floods resulted in 15 fatalities and the French Aude 2018 flash floods killed 13 people.

Economic losses followed a similar gradient—the Valencia disaster's estimated cost (EUR 3.5 billion insured, ~10.6 billion total losses) is orders of magnitude higher than, for example, the EUR 495 million insured losses in the Italy 2023 flood [65]. Even Greece's Thessaly event, which devastated one of the country's agricultural heartlands, had direct damage estimates around EUR 2 billion [66], lower than the Valencia losses. Several factors may explain these contrasts. Warning and evacuation efficacy were decisive factors: in Italy, authorities evacuated roughly 36,000 people ahead of the floods, undoubtedly reducing casualties. In Spain, despite meteorological warnings (upgraded to red alert on short notice), the speed and intensity of the Valencia floods overwhelmed many communities before evacuations could be completed. Urban planning and building practices also differed—for instance, Greek villages in Thessaly are generally low-rise and spread-out, whereas some Spanish localities had high population density in floodplain apartments.

These comparisons highlight that, although Mediterranean regions share a high exposure to extreme rainfall events, outcomes can vary widely. Crucially, preparedness and response measures (early warning systems, evacuation protocols) and the distribution of exposed assets (e.g., large cities vs. small towns, intensive agriculture vs. natural floodplains) strongly influence the severity of impacts. The Valencia event underscores the worst-case scenario of a cut-off low affecting a densely populated floodplain with limited preparedness, whereas the Italy and France cases show that even with substantial flooding, effective emergency management can lower the death toll. Learning from each event—Greek, Italian, French, and Spanish—is important to improve flood risk governance across the Mediterranean. In addition to these international events, the October 2024 flood can also be compared with historically significant floods in the Valencia region, particularly those of 1957 and 1982. While this study does not include a quantitative hydrometeorological comparison with these past events, the spatial extent, socio-economic disruption, and death toll observed in 2024 place the event among the most severe disasters in the recent history of the region. Further research incorporating historical cartographic archives and rainfall records would be valuable to contextualize the 2024 event within a long-term flood hazard framework.

In addition to comparing the October 2024 flood with other Mediterranean events, it is also relevant to contextualize our findings to recent work focusing on the same event. Alcaras (2025) [24] applied the Flood Mud Index (FMI), a novel spectral index based on red and blue Landsat 8 bands, to identify sediment-laden floodwaters in the Valencia region. Although no explicit flood extent in square kilometers is reported in that study, a visual comparison of the classification maps suggests a more conservative delineation of flooded areas compared to our multi-sensor mapping approach. This difference is likely attributable to various factors, including spectral indices applied (FMI vs. MNDWI) and classification thresholds. Our mapping results' comparative magnitude and spatial coverage suggest that the methodology employed here, although similarly based on Landsat 8, captured a more expansive and socio-economically critical representation of the event.

4.3. Limitations and Future Research

While this study demonstrated a successful application of remote sensing and GIS for flood impact assessment, it is important to recognize limitations that temper the findings and suggest avenues for future research.

First, the spatial and temporal resolution of the satellite imagery used constrains the analysis. Working at 10 m resolution limits the ability to detect highly localized flooding, such as inundation on small urban streets, inside buildings, or in narrow irrigation channels. Fine-grained flood dynamics in dense urban neighborhoods may not be fully captured when pixels aggregate over larger areas, potentially leading to an underestimation of impacts. Consequently, the flood extent maps likely underestimate minor or patchy flooding that could cause significant local damage. Increasing image resolution (using, for example, commercial satellites or UAV—Unmanned Aerial Vehicle aerial photos) could help resolve these details, though often at the expense of coverage area or temporal availability. A related issue is the temporal frequency of observations. The flood mapping in this study relied on essentially one-pass imagery from each sensor. However, floods are dynamic events and water levels may rise or recede between satellite overpasses. More frequent monitoring or integrating hydrological models to interpolate between observations would yield a more complete picture of the flood evolution over time.

Second, the resolution and type of ancillary data used for exposure analysis also introduce limitations. While suitable for national and regional-scale assessments, the WorldPop dataset used for population exposure estimates has ~1 km resolution. That dataset may overlook local variations in population density, especially in heterogeneous peri-urban settings. Similarly, the study did not incorporate real-time rainfall data, runoff modeling, or information on stormwater infrastructure—factors critical to understanding urban flood dynamics and the persistence of inundation. Future research should seek to incorporate finer-scale demographic data, municipal-level records, and urban drainage models to refine vulnerability and exposure assessments.

Third, no formal uncertainty or sensitivity analysis was conducted. While the combined use of SAR and optical data aimed to mitigate common limitations such as cloud cover or radar distortion, explicit uncertainty quantification remains challenging. In this study's case, the complex weather conditions and the area's characteristics made the generation of a benchmark for accuracy quantifications challenging as well as the evaluation of the corrections performed. Nevertheless, the internal consistency of the results across different sensors and their agreement with external products, such as the CEMS rapid mapping layer, support the robustness of the delineated flood extent.

Lastly, recent advances in automated image analysis techniques offer promising future directions. Machine learning classifiers, deep learning algorithms, and advanced change detection techniques have shown increasing potential to enhance flood mapping accuracy, particularly in complex urban environments where SAR data may be affected by speckle noise and heterogeneous backscatter [67,68]. Although not applied here, such methods could be integrated into future versions of this framework, especially for operational or near-real-time applications.

While the approach developed in this study offers a robust and scalable tool for post-event flood impact assessment, it remains best suited for spatially explicit, direct impact estimation. Further research that incorporates real-time hydrological data, finer-resolution exposure datasets, and automated classification techniques would enhance the applicability and precision of this type of analysis in support of flood risk governance.

4.4. Policy and Management Recommendations

The findings of this study carry several important implications for disaster risk management in Mediterranean urban areas like Valencia. In particular, they highlight the need for improved urban planning, infrastructure resilience, and incorporation of remote sensing into flood preparedness and response.

Integrate Flood Risk into Urban Planning and Land Use: Urban development policies should incorporate up-to-date flood hazard maps and historical flood data to guide where and how expansion occurs. The severe impacts observed in peri-urban Valencia—largely a product of buildings in low-lying floodplain areas—underscore that zoning regulations must restrict new construction in high-risk zones or mandate flood-proof building designs. Planners should preserve and even restore natural floodplains as buffer zones that can safely absorb floodwaters. Such nature-based solutions can reduce flood peaks while providing ecological and recreational benefits. This concept aligns with the scientifically consolidated approach of the "river territory", which advocates preserving fluvial corridors to reduce flood risks and enhance ecological functionality [69]. Such nature-based solutions can reduce flood peaks while providing ecological and recreational benefits.

Enhance Infrastructure Resilience and Protection: Critical infrastructure and utilities in known flood zones should be fortified and made resilient against flooding. Transportation infrastructure also requires attention: key road and rail corridors (like the A-7/AP-7 highways and regional rail lines) could be elevated in the most flood-prone stretches or supplemented by alternative routes so that the region is not cut off when a single corridor floods.

Improve Agricultural Land Management for Flood Mitigation: Given that agriculture was heavily affected and also plays a role in the region's hydrology, land management practices in rural areas should be part of the solution. Farmers and water authorities could work together to create controlled flood retention areas on certain farmlands to divert excess water away from populated areas during extreme events. Strengthening irrigation canals, embankments, and pumping stations in the Albufera rice district may help manage floodwaters and protect crops when feasible. Additionally, promoting agricultural insurance and fast compensation mechanisms would help rural communities recover faster after events, reducing long-term socio-economic harm. Integrating agricultural zones as active components of flood management strategies aligns with recent efforts to couple environmental restoration with socio-economic resilience, as highlighted in the Ebro River resilience framework [70]. By integrating agriculture into flood risk management (rather than viewing it as only a victim), the region can protect livelihoods and gain extra capacity to store floodwater in emergency scenarios.

Strengthen Early Warning Systems and Emergency Preparedness: Despite some official warnings, the high death toll in the Valencia event indicates a need to bolster early warning and response protocols. Meteorological agencies should continue to invest in advanced forecasting for cut-off lows and extreme Mediterranean rainfall; improving prediction lead times and accuracy will directly aid preparedness. Crucially, once warnings are issued, they must rapidly reach local authorities and the public through effective communication channels (sirens, mobile alerts, media broadcasts). Evacuation plans for neighborhoods within the flood risk areas should be established and regularly practiced so that residents know where to go when severe flood alerts are given. Therefore, investing in a robust flood early warning system is important. That includes installing upstream rain gauges and river level sensors, real-time monitoring networks, and protocols for inter-agency coordination (meteorological, civil protection, police, etc.). The experience from Italy's 2023 floods—where timely evacuations of tens of thousands were conducted—demonstrates that prepared communities can act on warnings to reduce significantly harm. Moreover, recent European frameworks emphasize the importance of integrating local communities and stakeholders in early warning systems to enhance territorial resilience [71]. Valencia and similar cities should review and update their emergency response plans to incorporate lessons from these events, ensuring that the response is swift and effective when extreme weather strikes.

Leverage Remote Sensing and GIS for Disaster Response and Recovery: The use of satellite data in this study exemplifies how remote sensing can be integrated into flood response. Emergency management agencies should establish procedures to obtain and use satellite-based flood maps in near real-time. We recommend developing an operational pipeline where Sentinel-1 SAR data, for example, is automatically processed within hours of acquisition to produce preliminary flood maps that civil protection units can use during the crisis. Drones and aerial surveys can complement this by providing high-resolution images of critical spots to direct urgent interventions. In the recovery phase, remote sensing and GIS should also be used to document damage and plan reconstruction. Mapping the flooded infrastructure allows officials to prioritize upgrades in those locations and to update flood risk models for the future. Overall, integrating remote sensing into the disaster management workflow—from early warning (satellite rainfall estimation, soil moisture monitoring) to response (flood extent mapping) and recovery (damage assessment)—will significantly strengthen resilience.

By adopting these recommendations, Mediterranean urban regions like Valencia can move toward a more proactive stance on flood risk, reducing potential losses from the kind of extreme event documented in this study.

5. Conclusions

This study presented a detailed spatial analysis of the October 2024 cut-off low flooding event in the Valencia Metropolitan Area of by integrating multi-sensor satellite data with socio-economic and infrastructure information. The main contributions and findings of this research can be summarized as follows:

- The developed workflow effectively combined Sentinel-1 SAR data, Sentinel-2, and Landsat 8 optical imagery, and high-resolution socio-economic datasets to accurately map flood extent and assess exposure across the metropolitan region.
- A total flooded area of approximately 199 km^2 was identified, with the most severely affected zones located in the southern and western peri-urban sectors, particularly in low-lying agricultural and residential areas.
- Critical infrastructure was significantly impacted, including portions of the primary road network, public transportation hubs, and health centers; exposure analysis revealed that around 7.8% of the metropolitan population resided in inundated areas.
- The integration of flood mapping with population density and land use data allowed the identification of vulnerability hotspots, illustrating the compounded risk faced by highly urbanized and economically active zones.
- The proposed methodology demonstrates the value of combining radar and optical satellite observations to enhance flood detection under adverse weather conditions, offering a transferable and scalable framework for flood impact assessment in Mediterranean urban contexts.
- Findings underscore the urgent need to incorporate detailed spatial exposure assessments into urban planning and flood mitigation strategies to enhance resilience against increasingly frequent and intense extreme weather events.

Overall, this study contributes to advancing flood risk analysis by providing a replicable approach that bridges remote sensing technologies with socio-economic vulnerability assessment at the metropolitan scale, offering practical insights for disaster risk reduction in densely populated coastal regions.

Author Contributions: Conceptualization, I.C.-M. and I.P.; methodology, I.C.-M., T.F., and E.B.; software, I.C.-M., T.F., and E.B.; validation, I.C.-M., T.F., and I.P.; formal analysis, I.C.-M., T.F., and E.B.; investigation, T.F. and E.B.; resources, T.F. and E.B.; data curation, I.C.-M., T.F., and E.B.; writing—original draft preparation, I.C.-M., T.F., and E.B.; writing—review and editing, I.C.-M., T.F., E.B., and I.P.; visualization, I.C.-M., T.F., and E.B.; supervision, I.C.-M. and I.P.; project administration, I.C.-M. and I.P. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: All data and results of this study are available from the corresponding author upon request.

Acknowledgments: The Department of Geography, Harokopio University of Athens, provided the required facilities for this study, which the authors gratefully acknowledge. The authors are grateful to the European Space Agency for providing Sentinel data.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AEMET	Spanish Meteorological Agency
AOI	Area of Interest
BOA	Bottom-of-Atmosphere
CEMS	Copernicus Emergency Management Service
CLC	CORINE Land Cover
CLMS	Copernicus Land Monitoring Service
DANA	Depresión Aislada en Niveles Altos
DEM	Digital Elevation Model
ESA	European Space Agency
FMI	Flood Mud Index
GIS	Geographic Information System
GRD	Ground Range Detected
IW	Interferometric Wide Swath
L2A	Level-2 A
MMU	Minimum Mapping Unit
MMW	Mapping Minimum Width
MNDWI	Modified Normalized Difference Water Index
MSI	MultiSpectral Imager
NASA	National Aeronautics and Space Administration
NDWI	Normalized Difference Water Index
NIR	Near Infrared
OLI	Operational Land Imager
OSM	OpenStreetMap
SAR	Synthetic Aperture Radar
SRTM	Shuttle Radar Topography Mission
SWIR	Short-Wave Infrared
TIRS	Thermal InfraRed Sensor
UAV	Unmanned Aerial Vehicle
USGS	United States Geological Survey
VV	Vertical–Vertical

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